



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma Green Belt (CLSSGB) Training

TGS Ref No: TGS-2025055775

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release - CLSSGB lesson plan.	Dr. Alfred Ang
2	19 July 2026	Major revision: rebuilt as a 4-day Green Belt progression from the Yellow Belt (CLSSYB). Content deepened to the CSSC Green Belt scope (Chapters 1-24) - adds sampling and sample size calculation, Measurement System Analysis and Gage R&R, hypothesis testing with p-values and Type I/II errors, correlation and regression, FMEA and RPN, Design of Experiments, SPC control chart selection with the eight out-of-control rules, and process capability Cp/Cpk. Expanded from 10 to 25 hands-on labs across the five DMAIC phases; teaching diagrams imported from the original v21 trainer deck; schedule and labs realigned to the v6 WA (SAQ) and PP assessment papers.	Dr. Alfred Ang
3	19 July 2026	Removed the Practice Exam slide and its asset - there is no Six Sigma practice exam on exams.tertiaryinfotech.com.	Dr. Alfred Ang
4	23 September 2026	Added the SIPOC & Process Map Builder (alfredang.github.io/sipoc) to the toolkit - wired into Lab 6 (SIPOC build, pain points, 'Check my SIPOC' validation) and Lab 7 (swimlane and handoff table), the Define phase slides, and a new Your	Dr. Alfred Ang

		Interactive Toolkit slide that introduces all five browser tools.	
5	23 September 2026	Restructured the labs so each lab owns a folder (labs/lab-NN-<name>/) containing its lab sheet and a data/ folder of Excel workbooks. Added 28 mock datasets generated from one reconciled month of Northwind order-fulfilment data - 4,200 orders with 357 late (8.5%), the exact baseline quoted in the WA (SAQ) and PP assessment papers - so every figure a learner calculates reconciles across the labs, the slides, the Learner Guide and the assessment. Lab steps now name the workbook and columns they operate on instead of referring to unspecified data.	Dr. Alfred Ang

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Course Information

Course Title	Certified Lean Six Sigma Green Belt (CLSSGB) Training
WSQ Course Reference	TGS-2025055775
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days · 8 training hours per day (32 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the Northwind Retail Distribution Centre improvement scenario
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Lead a Lean Six Sigma project — define the problem, scope the work and charter the project (A1, A2).
- LO2: Map and baseline a process using SIPOC, detailed process maps and value stream maps (A2).
- LO3: Build a valid measurement system — sampling, sample size and MSA/Gage R&R — and baseline capability (A4).
- LO4: Analyse process data statistically using Pareto, run charts, hypothesis testing, correlation and regression (A3).
- LO5: Identify and prove root causes of variation using Fishbone, 5 Whys and statistical evidence (A3, K2).
- LO6: Select, risk-assess and pilot improvements using solution selection matrices, FMEA and DOE (A5).
- LO7: Sustain the gain with SPC control charts, process capability and a control plan (A5, K1).

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 2 questions (K1, K2), 60 minutes, open book.
- Practical Performance (PP) — 3 applied DMAIC tasks (A1-A5), 90 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 4 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Six Sigma foundations, the Green Belt role and the Define phase

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM) [slides 2–19]
10:00–11:15	75 min	FOUNDATIONS — What is Quality; Lean vs Six Sigma vs Lean Six Sigma; the five Lean principles; history of Six Sigma; $Y = f(X)$; classical vs Six Sigma quality [slides 20–58]
11:15–11:30	15 min	Tea break
11:30–12:30	60 min	FOUNDATIONS — sigma level and DPMO; the 1.5 sigma shift; Cost of Poor Quality; belt roles and what a Green Belt leads; project selection and the cause/solution matrix; the DMAIC roadmap and tollgates
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 1: The Green Belt Role, Project Selection and Business Case; Lab 2: $Y = f(X)$, Sigma Level, DPMO and the DMAIC Roadmap [slides 59–66]
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	DMAIC · DEFINE — Voice of the Customer; affinity diagrams; Kano analysis; CTQ trees and operational definitions; the project charter; problem and goal statements; scope; SIPOC; stakeholders and RACI [slides 67–121]
16:45–18:00	75 min	Hands-on: Lab 3: Voice of the Customer, Affinity Diagram and Kano Analysis; Lab 4: CTQ Tree — Translating Customer Needs into Measurable Requirements [slides 122–129]
18:00–18:30	30 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Measure — process mapping, sampling, MSA and baseline capability

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory

9:45–11:00	75 min	digital attendance (AM) Hands-on: Lab 5: Project Charter, Problem Statement, Goal Statement and Scope; Lab 6: SIPOC, Stakeholder Analysis and RACI [slides 130–139]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · MEASURE — process mapping symbols and swimlanes; handoff analysis; the eight wastes (DOWNTIME); value-added analysis; value stream mapping; lean metrics and takt time [slides 140–214]
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Lab 7: Detailed Process Mapping and Swimlane Analysis; Lab 8: Value Stream Mapping, Takt Time and the Eight Wastes [slides 215–224]
15:00–15:15	15 min	Tea break
15:15–16:30	75 min	DMAIC · MEASURE — types of data; data collection plans; sampling techniques; sample size calculation; Measurement System Analysis and Gage R&R; yield, DPU, DPO, DPMO and RTY; descriptive statistics and baseline capability
16:30–18:15	105 min	Hands-on: Lab 9: Data Types, Operational Definitions and the Data Collection Plan; Lab 10: Sampling Techniques and Sample Size Calculation; Lab 11: Measurement System Analysis and Gage R&R [slides 225–238]
18:15–18:30	15 min	Day 2 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Analyze — root cause, hypothesis testing, correlation and regression

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:00	75 min	Hands-on: Lab 12: Yield, DPU, DPO, DPMO, RTY and the Hidden Factory; Lab 13: Descriptive Statistics, Normality and Baseline Process Capability [slides 239–249]
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	DMAIC · ANALYZE —

		common vs special cause variation; tampering; run charts and the six non-random patterns; the Pareto principle; stratification; boxplots [slides 250–320]
12:30–13:30	60 min	Lunch break
13:30–14:45	75 min	Hands-on: Lab 14: Variation, Run Charts and Stability Analysis; Lab 15: Pareto Analysis, Stratification and Boxplots [slides 321–330]
14:45–15:00	15 min	Tea break
15:00–16:30	90 min	DMAIC · ANALYZE — Fishbone (5M+E); 5 Whys; multi-voting; hypothesis testing (H_0/H_a, alpha, Type I and II errors, test selection, p-value decision rule); correlation and the coefficient of determination; regression
16:30–18:15	105 min	Hands-on: Lab 16: Fishbone, 5 Whys, Multi-Voting and Cause Prioritisation; Lab 17: Hypothesis Testing — Test Selection, p-values and Conclusions; Lab 18: Correlation, Regression and Quantifying the X-Y Relationship [slides 331–348]
18:15–18:30	15 min	Day 3 recap and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 4 — Improve, Control and the final assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)
9:45–10:45	60 min	DMAIC · IMPROVE — solution generation and brainwriting; benchmarking; the solution selection matrix; 5S; poka-yoke; pull, JIT and Heijunka; standard work; FMEA and RPN; Design of Experiments; piloting [slides 349–395]
10:45–11:00	15 min	Tea break
11:00–12:30	90 min	Hands-on: Lab 19: Solution Generation, Benchmarking and Brainwriting; Lab 20: Lean Countermeasures — 5S, Poka-Yoke, Pull, JIT and Standard Work; Lab 21: Solution Selection Matrix and Cost-Benefit Analysis; Lab 22: FMEA, Risk Priority Numbers,

		DOE and Piloting [slides 396–417]
12:30–13:30	60 min	Lunch break
13:30–14:30	60 min	DMAIC · CONTROL — process capability Cp and Cpk; control limits vs specification limits; SPC and control chart selection; the eight out-of-control rules; the control plan; SOPs; visual management; verifying the gain and handover [slides 418–469]
14:30–15:15	45 min	Hands-on: Lab 23: Statistical Process Control — Chart Selection and Control Limits [slides 470–476]
15:15–15:30	15 min	Tea break
15:30–15:45	15 min	Hands-on: Lab 24: Control Plan, SOP, Visual Management and Response Plan; Lab 25: Verify the Gain, A3 Storyboard, Handover and Project Closure [slides 477–489]
15:45–16:00	15 min	Course recap and Briefing for Assessment [slides 490–501]
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 2 questions, 60 minutes, open book
17:00–18:30	90 min	Practical Performance (PP) — 3 applied DMAIC tasks, 90 minutes, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: Six Sigma Foundations & The Green Belt Role	10%	Lab 1, Lab 2
DEFINE: Define — Scope the Problem	18%	Lab 3, Lab 4, Lab 5, Lab 6
MEASURE: Measure — Quantify Performance	24%	Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13
ANALYZE: Analyze — Find and Prove the Root Cause	26%	Lab 14, Lab 15, Lab 16, Lab 17, Lab 18
IMPROVE: Improve — Select, De-Risk and Pilot the Fix	12%	Lab 19, Lab 20, Lab 21, Lab 22
CONTROL: Control — Hold the Gain	10%	Lab 23, Lab 24, Lab 25